

You Say You Want a (Rose) Revolution? The Effects of Georgia's 2004 Market Reforms

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Abstract:

After Georgia declared its independence from the Soviet Union, it experienced civil war and strife for more than a decade. In late 2003, the peaceful Rose Revolution installed a new government that began a series of radical market-liberal reforms. However, the effectiveness of these reforms controversial. We offer a rigorous evaluation of these reforms via the synthetic control method which creates a credible counterfactual. Compared to the synthetic controls, we find that the reforms enacted after the Rose Revolution led to significant improvements in the Georgia's social and economic development, albeit with temporary side effects, and argue that this case provides some support for the effectiveness of rapid, multidimensional reform.

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I. Introduction

In late 2003, after more than a decade of civil war, political uncertainty, and economic collapse, the Rose Revolution peacefully deposed Eduard Shevardnadze, the former Soviet official who had ruled autocratically for most of the post-Soviet period.¹ In January 2004, a Columbia University-educated lawyer, Mikheil Saakashvili, was elected to lead the new government, and Kakha Bendukidze was tapped by Saakashvili to lead the reform process. Although the issues related to separatist regions and hostile relations with Russia remained, the Rose Revolution ultimately ushered in a period of rapid economic liberalization.

The purpose of this study is to rigorously evaluate the effect of these reforms. In order to do this, we need a reasonable counterfactual. That is, we need to know what would have happened in Georgia absent the reforms. To that end, we employ the Synthetic Control Method (SCM) that allows us to compare Georgia's socio-economic performance after the Rose Revolution against a weighted average of comparable countries with similar characteristics.

We find large positive and significant effects of reform on incomes and large negative and significant effects on infant mortality relative to our statistical counterfactual. Inequality increases and employment declines and then recovers but the effects are not statistically significant when compared to the time path of our synthetic control. We conclude that the Rose Revolution on balance was quite positive for the country of Georgia.

In what follows below, we provide a brief overview of Georgia's experience with liberalization and discuss the expected effects the reforms, explain how the SCM method

¹ See Burakova and Lawson (2013) for a more detailed treatment of the political economy of Georgia.

works and discuss identification, and then present our results on the effect of the Rose Revolution on per-capita income, infant mortality, income inequality, and employment.

II. Liberalization in Georgia

Georgia's market-liberal reforms earned the country a reputation as one of the fastest-reforming nations in the former Soviet bloc, if not the entire world. Simeon Djankov, founder of the World Bank's Doing Business project and former minister of finance of Bulgaria, said Georgia's reforms were "unprecedented," (USAID, 2009: 43) with the closest analogues being Singapore in the 1960s, South Korea in the 1970s, Ireland in the 1980s, and Estonia in the 1990s. USAID (2009: 3) claims Georgia has made "the broadest, deepest, fastest business climate reforms of any country in the last fifty years."

Evidence of the rapid pace of reform can be found in various measures of the policy and institutional environment. Figure 1 shows the 2004-2014 ratings for Georgia for the Economic Freedom of the World index (Gwartney, Lawson, and Hall, 2016), the Doing Business rating for enforcing contracts (World Bank, 2017), and the Corruption Perception Index (Transparency International, 2016). Each data series was normalized to zero in 2004, and the data are expressed in standardized units. The Economic Freedom of the World rating increased by 0.90 standard units. The rating for enforcing contracts improved by 0.74. Finally, the Corruption Perception Index improved by an impressive 1.46 standard units. All three indicators agree that Georgia has experienced significant improvement in its policy environment.

The Rose Revolution ushered in a plethora of individual reforms, and since their implementation was roughly simultaneous, we cannot hope to distinguish the effect of

individual measures.² In our view, the four main elements of Georgia's reform package were anti-corruption, privatization, tax reform, and liberalization. Even though we are measuring the treatment effect of the entire package, it is worthwhile to briefly consider the economics linking these four main policy groupings to economic outcomes.

The consensus view on corruption is that reducing it improves economic performance by removing distortions (Mauro (1995)). However, this picture has been shown to be more nuanced than what was reported in these initial papers with researchers showing the effect of corruption on growth to be non-linear, conditional on the form of government, and conditional on the type of corruption.³ Given the high baseline levels of corruption pre-Rose Revolution, we would expect the (highly effective) anti-corruption campaign to have improved economic performance in Georgia.

The macroeconomic impact of privatization is a contentious issue. Megginson and Netter (2001) has shown in developed countries that it increases efficiency on the micro level, which should in turn raise national income. There is less literature on the macro effects of privatization, but Bennett et al (2007) argue that the effect depends on the type of privatization employed. Also relevant here is the large literature on whether the reform is done quickly (big bang reform) or gradually. The consensus is that gradual reform is better

² Burakova and Lawson (2013) list and describe 28 distinct reform initiatives undertaken between 2004 and 2007 during the heyday of the reform process under Saakashvili's and Bendukidze's leadership. Major tax reforms included both reducing the levels and number of taxes. The income tax was initially set at a flat 12% but was later combined with the payroll tax at a 20% flat rate. Privatization of vast amounts of state-owned assets was achieved mainly through open and transparent highest-bidder auctions with very little apparent favoritism. The new labor code was only a few pages long and offered few guarantees to workers; unions, though perfectly legal, were given no special legal status – this of course was a major departure from the old Soviet system. Interestingly, Georgia abolished anti-monopoly laws while simultaneously opening to external trade as most goods can be imported tariff free. So-called “de-bureaucratization” efforts resulted in 95% reductions in executive branch staffing and the elimination of 6 entire ministries and 18 agencies. To combat corruption at the local level, Saakashvili's government shocked the nation by firing the nation's entire police force, some 30,000 officers.

³ For a nuanced view of the effects of corruption see Pavlik (2018). On how the effect of corruption is conditioned by form of government, see Drury et al (2006).

for economic performance. Georgia's privatizations were fairly transparent, but very fast, so the expected effect short run effect in isolation is perhaps ambiguous.

Tax reform is expected to improve economic performance much in the way reducing corruption does, by removing distortions and disincentives. However, unlike corruption, which might facilitate economic activity in some circumstances, tax reform seems to be a clear positive for GDP.⁴ Note that Georgia was able to combine tax reform with large increases in revenue collected, so potential negative indirect effects of reform via lower government revenues are not relevant here.

Liberalization includes the so-called de-bureaucratization efforts to reduce staffing within the executive ministries, labor market reform, and the elimination or streamlining of business permits. Again, the intended effect of these reforms is to encourage entrepreneurship, reduce distortions and raise efficiency, all of which should increase economic activity. However, government regulations that establish and enforce the "rules of the game" are requisites for the efficient operation of markets, so reforms of this type clearly risk going too far and actually retarding economic activity. However, as was the case with corruption, in light of Georgia's extraordinarily high regulatory starting point, these market liberalizations could be plausibly expected to improve economic efficiency and thus GDP.

To date, there is virtually no scholarly literature on the economic effects of the Rose Revolution or the market liberal reforms that ensued. Papava (2005, 2013, and 2014) does offer highly critical accounts of the post-Rose Revolution period, paradoxically calling it "a symbiosis of neoliberalism and neobolshevism" (Papava, 2013: 51), but there is essentially no data analysis in these papers. Most of the rest of the scholarly literature deals with the

⁴ See Lee & Gordon (2005) for a study of the effect of corporate tax cuts (which Georgia employed) on growth.

political and international relations situation (see for example, Mitchell, 2008) and is similarly anecdotal.⁵

The sheer scale and scope of these reforms and the speed with which they were implemented was quite astounding. The question is, did they matter, and if so, how much? That is, were these rapid multidimensional reforms, “shock treatment” if you will, a good thing for the country?

III. Creating the Synthetic Control

We want to estimate the impact of the market-liberal reforms unleashed by the Rose Revolution in Georgia on various socio-economic outcomes. Ideally, we would like to observe what happened to Georgia in the absence of the reforms. Of course, this is impossible as there is only one Georgia and it experienced the reforms, and there is no Georgia that did not experience the reforms. Instead, we use pre-revolution data to create a “synthetic Georgia”, a weighted average of control countries similar to Georgia. This synthetic control is designed to track pre-revolution Georgian outcomes and also to match actual Georgia on the values of several indicator variables as well. To the extent the synthetic Georgia accurately represents the actual Georgia, it can serve as the counterfactual we need.

The synthetic control literature

Abadie and Gardeazabal (2003) developed the synthetic control method in order to identify the effect of terrorism in the Basque Country in Spain. Abadie et al. (2010) explain the technique in detail and analyze the effect of a change in California's tobacco regulations.

⁵ Tangiashvili and Slade (2014) do offer an interesting and quite critical data-driven analysis of Georgia's zero-tolerance policies related to school violence in Saakashvili's two terms in office.

Abadie et al. (2015) further refine the method and apply it to the economic effects of German reunification. Gautier et al. (2009) and Montalvo (2011) provide additional applications to the analysis of terrorism. Synthetic control methods have also been used to study affirmative action (Hinrichs, 2012), compulsory voting (Fowler, 2013), economic liberalization (Billmeier and Nannicini, 2013), natural disasters Cavallo et al. (2013), and the impact of Hugo Chavez in Venezuela (Grier and Maynard, 2016).

The synthetic control method

The most important aspects of the method are selecting the countries, the indicator variables, and what weights to assign to indicators and the countries comprising the synthetic Georgia. Our goal is to create a synthetic version of Georgia from a list of potentially similar countries that mimics the conditions in Georgia leading up to the Rose Revolution. Thus, in order to create the synthetic control, we need to find a set of countries and country-weights that minimizes the pre-Rose Revolution differences in socio-economic conditions between actual Georgia and the synthetic control Georgia.

In addition to similar socio-economic conditions leading up to 2004, we also want our synthetic control countries to be economically and structurally similar to Georgia. To that end, we also use various indicator variables to minimize the difference between the weighted average of these variables for the synthetic control and Georgia. In the final analysis, indicators with more predictive power for the outcome variable and countries with more similar values of the more relevant predictors will receive higher weights in creating the synthetic control.

Inference

One issue economists often have with case studies is the lack of significance tests for the results. Abadie et. al. (2010) address this issue by means of permutation testing.

Specifically, we want to see whether the deterioration in the ability of the control to match Georgia post 2004 is large relative to random deviations using the procedure where there was no intervention. That is, we conduct a synthetic control analysis as described above for all the countries in our control group. These tests are generally referred to as placebo tests. Since the other countries did not undergo the Rose Revolution in 2004, we would not expect to see significant changes in the predictability of their outcomes.

We use these placebo tests to create a unique p-value for each post-treatment period, allowing us to see how the significance of the estimated effect changes over time. To calculate the p-values for time i , we collect each placebo's estimated treatment effect for the i th period and divide them by their respective pre-treatment RMSPE. This division minimizes the potential problem of poor pre-treatment fit driving an estimated effect in a post-treatment period. Since we conduct a two-sided test, it is also necessary to calculate the absolute value of each effect. Once the statistics are collected, we generate period i 's p-value estimate by counting the number of placebos with greater effects than that of Georgia and divide by the number of total countries (donors and Georgia). The process is repeated for all post-treatment periods.⁶

Identification

All methods of causal inference require some identifying assumptions and synthetic control is no exception. There are two requirements for our results to be causal. The first is that the treatment does not affect any of the control countries. This is known as the stable unit treatment value assumption (SUTVA). While this is an assumption and not provable, we have shown in the direct sense that this is likely true as none of the countries in our donor

⁶ We use the “Synth_Runner” software (Galiano & Quistorff, 2016) to implement this significance testing in Stata.

pool experienced anything like the same level of reforms that Georgia did during this time period. However, to be certain SUTVA holds, we need to also rule out indirect effects (spillovers).⁷ It is worth noting though in our case, to the extent that SUTVA may be violated, it would tend to cause us to underestimate the effect of the reforms under study.

The second identifying assumption is that the effects we find are due only to the treatment that we claim. In other words, the assumption is that there are no other simultaneous or subsequent treatments that may be causing the results we see. The most likely violation of this assumption would be Russia's brief invasion of Georgia in 2008. However, the most likely effect of the invasion would be to reduce economic performance in Georgia, and that would again cause us to underestimate the true treatment effect of the Rose Revolution reforms.

Finally, it is worth noting that if these assumptions are satisfied and we can produce an acceptable control, then we can recover causal estimates of the effects of Georgia's reforms. However, the synthetic control method by design cannot distinguish between which reforms were most important or by what channels the reforms produced their impacts.

⁷ The most likely spillover would arise if reforms make Georgia grow faster and that would tend to increase Georgia's imports. If a control country was a major exporter to Georgia, it would also tend to see its GDP increase. This is very hard to completely rule out, but note that if it occurs, it would tend to raise the synthetic control and understate any true effects. As of 2016, the 10 largest exporters into Georgia are Turkey, Russia, China, Azerbaijan, Ukraine, Germany, Italy, the US, Japan, and Romania. Information comes from this source: <https://atlas.media.mit.edu/en/profile/country/geo/>. Turkey only gets a positive weight in our inequality model, Russia does not get any weight above 1% in any of our models, China is not in our donor pool, Azerbaijan only gets a positive weight in our employment model, Ukraine does not get any weight above 1% in any of our models, Germany, Italy, the US, and Japan are not in our donor pool, and finally Romania only gets a positive weight in our employment model. Again, we cannot ever prove SUTVA holds, but in our view, it is not badly violated in our experiments.

IV. The Experiments

We conduct four quasi-experiments in this paper. Each compares the post-Rose Revolution experience from 2004 onward of the actual Georgia with its synthetic Georgia counterpart. The outcome variables to be considered are per-capita GDP, infant mortality, income inequality, and the employment/population ratio. In each separate case, a synthetic Georgia will be constructed that mirrors the actual Georgia's pre-Rose Revolution record from 1995-2003 for that outcome variable.⁸ Table 1 provides descriptive statistics for both the indicator variables and the outcome variables used. The first column of Table 2 lists all the potential donor countries included in the study.⁹ The last four columns of Table 2 list the country weights generated by the synthetic control method to create a synthetic Georgia for each of the four experiments we consider.

Per-Capita GDP Results

The first experiment looks at per-capita GDP. Seven indicator variables along with two lags of per-capita GDP were used in constructing the control. There is a large literature on the empirical determinants of economic growth and from that literature we use Investment, Foreign Direct Investment, Exports, and Inflation. Ideally, we would like to include an education measure, but those data are just too sparse for Georgia and several of our other donor countries to allow it. We also add percent of the population living in urban

⁸ Data for Georgia begin in 1991, but a severe hyper-inflation in '92-'93 make it impossible for us to find a synthetic that looks at all like Georgia in terms of macroeconomic and social indicators for those years. Thus, we begin the analysis in 1995.

⁹ We do not claim that none of these countries experienced any type of economic reform. We are claiming that the extent and rapidity of Georgia's reform experience is unique in this group over the 2004-20014 period. Of all the countries in our list, Georgia's experience on the three reform indicators listed in the text is the largest. The next closest countries are Poland and Serbia. Interestingly, neither of these two countries make it into any of the estimated synthetic controls in the paper with a weight about 1%. To the extent that countries in the synthetics have experienced similar reforms to Georgia, our results on the impact of Georgia's reforms will be understated.

areas, life expectancy, and military spending as a share of GDP as additional socio-demographic variables to match on.

Our estimated synthetic Georgia is comprised of 23% Armenia, 21% Bosnia and Herzegovina, 54% Moldova, and 2% Tajikistan. When comparing Georgia's per-capita GDP with that of its synthetic control counterpart during the pre-Rose Revolution era, we find a RMSPE of 47.65. The last column of Table 3 shows that the donor pool average's ability to track actual Georgia is more than 10 times worse than that of our synthetic Georgia. These calculations highlight the value of creating a synthetic control using a weighted average of similar nations instead of using the average of all the countries in the donor pool. The covariate values for synthetic Georgia match fairly well, or at least better than do the values from the donor pool average, except for exports.¹⁰

As Figure 2 illustrates, the pre-Rose Revolution actual Georgia and synthetic Georgia performed very similarly, but after the Rose Revolution, per-capita GDP grew substantially faster in actual Georgia relative to its synthetic control. This suggests that the impact of the Rose Revolution and the market liberal reforms gave per-capita GDP a boost in Georgia over what would have happened otherwise. And not just a small boost: by the end of our study period actual income is over 33% higher than predicted by the control! And remember, the control was very accurate in predicting Georgian income before the revolution.

¹⁰ The method of creating the control that we use, following Abadie, involves solving for a set of weights that create the control along with values for a positive-definite matrix (called the V matrix), whose values indicate the importance of the associated covariate in determining the outcome variable being studied. Intuitively, the algorithm pushes the synthetic control's covariate values closer to the actual covariate values for the covariates that are most useful in predicting the outcome variable. The variables that do not match well have very small corresponding entries in the V matrix. For example, the export variable does not match well between actual and synthetic Georgia, but that variable's entry in the V matrix is tiny. When the V matrix entries are normalized to sum to one, exports' entry is virtually zero (0.00075).

Figure 3 presents significance tests for each year and shows that the large estimated differences between Georgia's actual post-Rose Revolution experience and its synthetic control were statistically significant at the 0.10 level or better in 8 of the twelve years we examine.¹¹

As a robustness check, we switch datasets from the World Bank data used in the body of the paper to the Penn World Tables (PWT) dataset and repeat our experiment on real per capita income using those data. The results are in the Appendix. We find very similar results. The effect sizes are quite similar, with the PWT results showing a slightly smaller peak effect at the end, but the significance levels with PWT data are slightly stronger (8 of the 12 periods significant at 0.05 or better).

Infant Mortality Results

Our second case is infant mortality. To construct the control, we use 6 indicator variables: percent urban, investment, life expectancy, inflation, per-capita GDP, and the gender ratio along with 3 lags of infant mortality. Synthetic Georgia is made up of 46% Armenia, 35% Moldova, 3% Slovenia, and 6% Tajikistan.

In this case, the values of the predictor variables for synthetic Georgia match up very well with those for actual Georgia, and are clearly better than the average values of those variables across the donor pool. The pre-Rose Revolution RMSPE of predicting Georgia by its synthetic control is much smaller than the RMSPE for predicting Georgia by the donor pool average (0.072 vs 3.35).

As Figure 4 illustrates, the pre-Rose Revolution actual Georgia and synthetic Georgia performed very similarly, but after the Rose Revolution infant mortality fell substantially

¹¹ It is true that the significance levels do not often hit 0.05, but the size of the effects coupled with their marginal significance means we should take them seriously. For a good discussion of economic vs. statistical significance, see Goldberger (1991) or McCloskey & Ziliak (1996).

faster in actual Georgia compared with its synthetic control. Infant mortality improved after the Rose Revolution faster than it would have without the intervention. Again, the effect is substantial. At the end of the period, infant mortality is almost 25% lower in Georgia compared to its synthetic control.

Figure 5 presents our significance tests by year for the effect of the Rose Revolution on infant mortality. 8 of the 12 treatment periods are significant at the 0.10 level or better and the other four all have p-values of 0.11. Our analysis shows that the observed reduction in infant mortality in Georgia was in fact attributable to the Rose Revolution and its policy reforms. The bottom line is that we see a large and significant improvement in infant mortality beginning in 2004 in Georgia, i.e., after the Rose Revolution relative to the prediction of the “business as usual” synthetic control.

Income Inequality Results

Liberalizations are often accused of sacrificing equity for growth. Thus, our third experiment is designed to assess this argument by examining income inequality as measured by the Gini coefficient provided by Solt (2016).¹² Here we use FDI, percentage of the population living in urban areas, the investment rate, and three lags of the Gini as our indicator variables. For this Gini coefficient experiment, synthetic Georgia is comprised of, 20% Angola, 2% Bosnia, 10% Czech Republic, 3% Jordan, 2% Kazakhstan, 2% Kyrgyz Republic, 6% Macedonia, 4%, Serbia 7% Sri Lanka, and 28% Turkey.¹³

¹² We use Stata’s “ipolate” command to fill in missing data. The command uses the following formula to estimate missing y values: $y = \frac{y_1 - y_0}{x_1 - x_0}(x - x_0) + y_0$, such that $x_0 < x$ and $x_1 > x$, where y_0 and y_1 are observed.

¹³ The percentages do not add up to 100 because several countries also enter with a weight of 1% each, specifically Algeria, Armenia, Azerbaijan, Estonia, Latvia, Lithuania, Macedonia, Moldova, Mongolia, Morocco, Philippines, Poland, Russia, Tunisia, Turkmenistan, Ukraine, and Uzbekistan

Again, the method does an excellent job of finding a synthetic version of Georgia that matches Georgia's income inequality before 2004. The RMSPE between Georgia and its synthetic control was just 0.192, which is much smaller than the RMPSE obtained using the donor pool average (5.53). Synthetic Georgia's values for our indicator variables also match up quite well with the values of actual Georgia.

Figure 6 shows that the pre-Rose Revolution actual Georgia and synthetic Georgia performed very similarly, but after the Rose Revolution the Gini increased in actual Georgia compared with its synthetic control. It appears income inequality worsened after the Rose Revolution faster than it would have without the intervention.

Figure 7 presents our significance tests, and shows that the effect is not significant at conventional levels in any of the years from 2004 on, though the 3 years with the largest effects all have p-values of 0.14. While inequality did not rise dramatically in post Rose Revolution Georgia, it fell dramatically in synthetic Georgia, indicating that the Revolution did exacerbate inequality, though the effect is not statistically significant.

Employment Results

The final experiment examines employment, specifically the employment/population ratio. We study employment because we know the Rose Revolution had a direct and dramatic impact on government employment. In a country of fewer than 5 million people, 95% reductions in various government ministries and the firing of tens of thousands of police officers seems likely to impact aggregate employment numbers. In this case our statistical procedure chooses a synthetic control comprised of Albania (21%), Czech Republic (1%) Kazakhstan (53%), Pakistan (1%), Poland (8%), Romania (10%), and Russia (1%), with 30 other donors contributing less than 1%. We use the age dependency ratio,

FDI, percent of the population in urban areas, inflation, and trade share in GDP along with the average value of the employment ratio as our indicators variables.

This synthetic control again has values of the indicators relatively close to the values for actual Georgia and greatly outperforms the donor pool average in mimicking Georgia's employment/population ratio leading up to 2004 (RMSPE of 0.84 vs 7.43).¹⁴ After 2004, we find a large initial drop in Georgian employment relative to its synthetic control and then a sharp rebound around 2009 as shown in figure 8. Figure 9 contains the year by year significance tests and shows the effect of the Rose Revolution on employment was not statistically significant.

Again, given the size of these effects, there is little question that Georgia's reforms caused short run labor market disruptions as employees in the government itself and in newly privatized state-owned enterprises lost their jobs. The good news is that, in this case, these workers had the opportunity to find new employment within a reasonable time frame so that the negative effects were temporary.

V. Conclusion

By all accounts, Georgia experienced one of the world's fastest and deepest economics liberalizations during the years immediately following its Rose Revolution in late 2003. However, it is incorrect to try and measure the effect of the reforms simply by looking at gross changes in Georgia's socio-economic performance. The relevant comparison is to what would have happened if in Georgia if the reforms had not occurred.

¹⁴ As in the GDP case, the synthetic control for employment does not match the trade variable very well. In this case as well, the trade variable has by far the smallest value in the V matrix, meaning it is not very important for predicting employment outcomes. In pre-Rose revolution Georgia, trade was not an important factor in determining output and employment.

This study uses the SCM to examine how Georgia performed compared to a synthetic control version of Georgia, a weighted average of similar countries, that did not experience the same reform process. The results suggest that the per-capita GDP and infant mortality improvements that Georgia experienced after the Rose Revolution were much greater than would have been expected in the absence of the reforms. In contrast, there is no evidence supporting the notion that the Rose Revolution's market-liberal reforms caused any increase in income inequality. Finally, the liberalizations do appear to have caused large employment losses in the years immediately following the Rose Revolution, though the dislocated workers were mostly able to find new employment by the end of the period perhaps thanks to the newly liberalized labor market.

In sum, the Rose Revolution generated large and significant increases in per-capita incomes (33%) along with large and significant declines in infant mortality (-25%) relative to the synthetic control, without significantly worsening inequality. The employment ratio initially fell, but recovered by the end of our period of study.

These findings should provide reform advocates with some hope. The Rose Revolution greatly improved economic and health outcomes in Georgia with fairly short-lasting negative employment side effects and no significant exacerbation of inequality. Georgia's experience shows that "big-bang" reform can work.

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Appendix: Using Penn World Tables Data to Assess the Effect of Georgia's Reforms on Real Per Capita Income

Here we switch datasets from the World Bank to the Penn World Tables, as a robustness check on the results for real per capita income that we found in the body of the paper. Table A1 reports the indicator variables we use, and shows that we achieve “good but not great” matching on these variables between actual and synthetic Georgia, though the matching is better for synthetic Georgia than for the donor pool average. Synthetic Georgia tracks actual Georgia very well in the pre-2003 period (RMSPE of 35 compared the donor pool average RMSPE of 5000). Figure A1 illustrates the close pre-treatment tracking and then shows the sharp divergence post-2003 where Georgia outperforms its synthetic control, just as we found in the body of the paper. Finally, figure A2 gives the size of the effect and the period by period significance levels. The estimated treatment effects are similarly sized to those reported in the paper, though slightly smaller at the beginning and end of the treatment period and slightly larger in the middle. 8 of the 12 periods are significant at the 0.05 level or better.

TABLE 1: SUMMARY STATISTICS

Variable	Mean	Standard Deviation	Observations	Description
GDP per Capita	\$11,257.67	(7,560.66)	861	GDP per capita, PPP (constant 2011 international \$)
Infant Mortality	23.15	(21.97)	882	Mortality rate, infant (per 1,000 live births)
Gini	34.14	(6.19)	659	Gini inequality index (SWIID)
FDI	4.53	(5.91)	868	Foreign direct investment, net inflows (% of GDP)
Employment Ratio	50.20	(8.56)	882	Employment to population ratio, 15+ total (%)
Urban	54.40	(16.36)	882	Urban population (% of total)
Investment	25.01	(7.11)	841	Gross capital formation (% of GDP)
Life Expectancy	71.01	(4.87)	878	Life expectancy at birth, total (years)
Military	2.55	(1.71)	789	Military expenditure (% of GDP)
Age Dependency	55.68	(13.75)	882	Age dependency ratio (% of working-age population)
Exports	40.92	(17.69)	866	Exports of goods and services (% of GDP)
Gender Ratio	48.96	(1.48)	882	Population, male (% of total)
Inflation	27.44	(188.08)	866	Inflation, GDP deflator (annual %)

Note. This table contains summary statistics for all variables in all analyses. The statistics are calculated using all countries, donors and Georgia, for all periods of the analyses, 1995-2015. There are 630 observations in the panel data. For the descriptions, we draw directly from the metadata of the respective sources. *Source.* World Bank's World Development Indicators, Standardized World Income Inequality (SWIID) and authors' calculations.

TABLE 2: ESTIMATED SYNTHETIC CONTROL WEIGHTS BY OUTCOME VARIABLE

Country	Income	Infant Mortality	Inequality	Employment
Albania	0.00	0.00	--	0.21
Algeria	0.00	0.00	0.01	0.00
Angola	0.00	0.00	0.20	0.00
Armenia	0.23	0.46	0.01	0.00
Azerbaijan	0.00	0.00	0.01	0.00
Belarus	0.00	0.00	0.00	0.00
Bosnia	0.21	0.00	0.02	0.00
Bulgaria	0.00	0.00	0.00	0.00
Croatia	0.00	0.00	0.00	0.00
Czech Republic	0.00	0.00	0.10	0.01
Egypt	0.00	0.00	0.00	0.00
Estonia	0.00	0.00	0.01	0.00
Hungary	0.00	0.00	0.00	0.00
Israel	0.00	0.00	0.00	0.00
Jordan	0.00	0.00	0.03	0.00
Kazakhstan	0.00	0.00	0.02	0.53
Kyrgyz Republic	0.00	0.00	0.02	0.00
Latvia	0.00	0.00	0.01	0.00
Lithuania	0.00	0.00	0.01	0.00
Macedonia	0.00	0.00	0.06	0.00
Moldova	0.54	0.35	0.01	0.00
Mongolia	0.00	0.00	0.01	0.00
Morocco	0.00	0.00	0.01	0.00
Pakistan	0.00	0.00	0.00	0.01
Philippines	0.00	0.00	0.01	0.00
Poland	0.00	0.00	0.01	0.08
Romania	0.00	0.00	0.00	0.10
Russia	0.00	0.00	0.01	0.01
Samoa	--	--	--	0.00
Serbia	0.00	0.00	0.04	0.01
Slovakia	0.00	0.00	0.00	0.00
Slovenia	0.00	0.03	0.00	0.00
Sri Lanka	0.00	0.00	0.07	0.00
Syrian Arab Republic	--	--	--	0.00
Tajikistan	0.02	0.06	0.00	0.00
Tonga	--	--	--	0.00
Tunisia	0.00	0.00	0.01	0.00
Turkey	0.00	0.00	0.28	0.00
Turkmenistan	0.00	0.00	0.01	0.00
Ukraine	0.00	0.00	0.01	0.00
Uzbekistan	0.00	0.10	0.01	0.00

Note. Columns show the estimated weight for the synthetic Georgia. Each column represents an outcome variable, labelled at the top of the column. Values are in percentage points. Donors that receive a positive weight are in bold for the reader to more easily identify. Values are rounded, so the columns may not sum to one.

TABLE 3: INCOME PREDICTOR MEANS

Variables	Actual Georgia	Synthetic Georgia	Donor Pool Avg.
GDP per Capita (1996)	\$2,616.97	\$2,615.55	\$7,889.09
GDP per Capita (2001)	\$3,445.58	\$3,476.57	\$9,295.86
FDI	5.43	4.18	3.67
Urban	52.94	48.38	53.53
Investment	23.54	23.27	23.72
Life Expectancy	71.56	69.16	69.23
Military	1.12	1.77	2.85
Inflation	27.60	19.79	57.58
Exports	31.84	43.94	40.61
RMSPE	--	47.649	5584.773

Note. This table shows the values of indicator variables for different comparison groups. In doing so, it illustrates the strenghts of the synthetic control, which better fits the behavior of actual Georgia in the pre-treatment period (1995-2004). Unless otherwise indicated, variables are averaged across the pre-treatment period. Please refer to table 1 for a description of the variables. The final row gives the root mean square prediction error for the unit of comparison.

TABLE 4: INFANT MORTALITY PREDICTOR MEANS

Variables	Actual Georgia	Synthetic Georgia	Donor Pool Avg.
Infant Mortality (1995)	37.70	37.67	33.51
Infant Mortality (2000)	31.00	30.92	27.96
Infant Mortality (2003)	25.50	25.59	24.52
Urban	52.94	52.99	53.53
Investment	23.54	21.62	23.72
Life Expectancy	71.56	68.90	69.23
Inflation	27.60	33.27	57.58
GDP per Capita	\$3,176.69	\$3,181.01	\$8,751.91
Gender Ratio	47.41	47.81	49.07
RMSPE	--	0.072	3.348

Note. This table shows the values of indicator variables for different comparison groups. In doing so, it illustrates the strenghts of the synthetic control, which better fits the behavior of actual Georgia in the pre-treatment period (1995-2004). Unless otherwise indicated, variables are averaged across the pre-treatment period. Please refer to table 1 for a description of the variables. The final row gives the root mean square prediction error for the unit of comparison.

TABLE 5: INEQUALITY PREDICTOR MEANS

Variables	Actual Georgia	Synthetic Georgia	Donor Pool Avg.
Gini (1995)	37.84	37.80	34.05
Gini (2000)	39.85	39.83	34.01
Gini (2003)	40.80	40.78	34.17
FDI	5.43	5.42	3.67
Urban	52.94	52.89	53.53
Investment	23.54	23.51	23.72
RMSPE	--	0.192	5.534

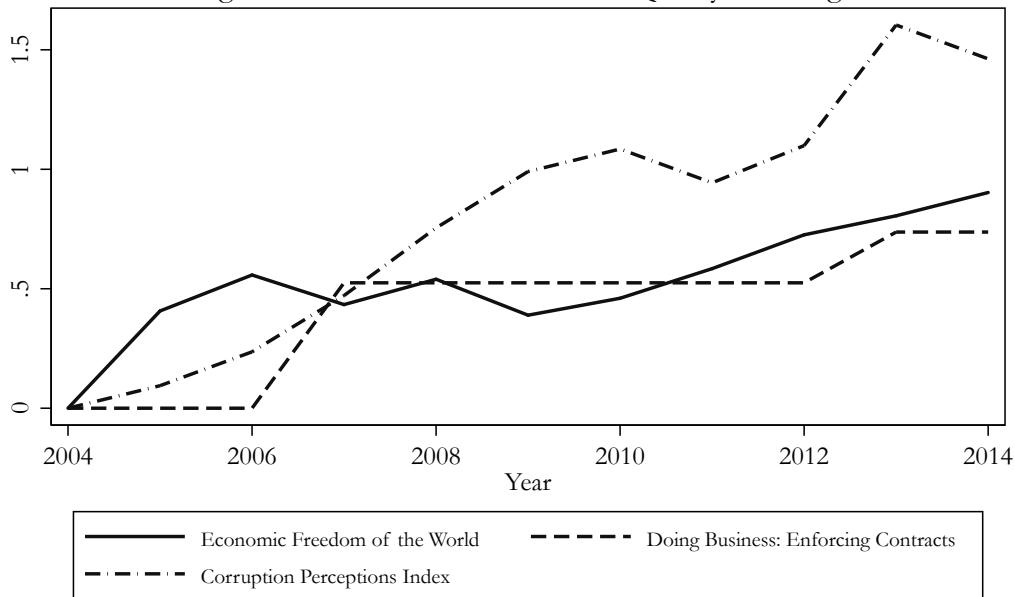
Note. This table shows the values of indicator variables for different comparison groups. In doing so, it illustrates the strenghts of the synthetic control, which better fits the behavior of actual Georgia in the pre-treatment period (1995-2004). Unless otherwise indicated, variables are averaged across the pre-treatment period. Please refer to table 1 for a description of the variables. The final row gives the root mean square prediction error for the unit of comparison.

TABLE 6: EMPLOYMENT RATIO PREDICTOR MEANS

Variables	Actual Georgia	Synthetic Georgia	Donor Pool Avg.
Employment Ratio	57.18	57.16	49.82
Age Dependency	54.44	54.45	60.02
FDI	5.43	5.42	3.67
Urban	52.94	52.99	53.53
Inflation	27.60	27.31	57.58
Trade	61.99	72.45	84.80
RMSPE	--	0.84	7.43

Note. This table shows the values of indicator variables for different comparison groups. In doing so, it illustrates the strenghts of the synthetic control, which better fits the behavior of actual Georgia in the pre-treatment period (1995-2004). Unless otherwise indicated, variables are averaged across the pre-treatment period. Please refer to table 1 for a description of the variables. The final row gives the root mean square prediction error for the unit of comparison.

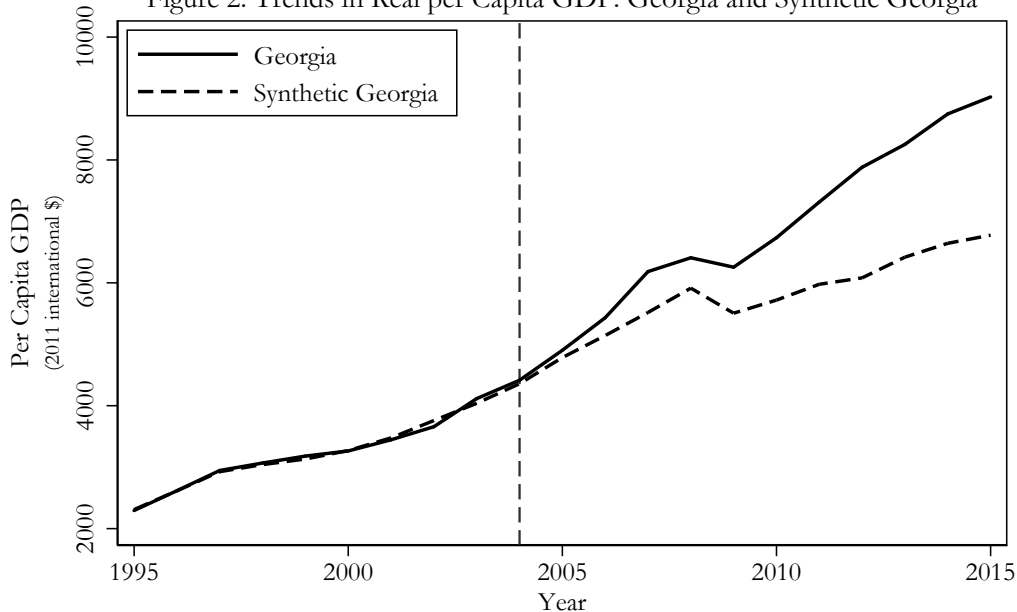
Figure 1: Measures of Institutional Quality in Georgia



Note. Following the Rose Revolution, Georgia's reform led to improvements in various institutional quality measures. This figure demonstrates the increases experienced by three different indexes, listed in the figure's key. Each data series begins with a starting value of zero, and the units are expressed in standardized terms. A one unit increase along the y-axis represents a one standard deviation in the respective index.

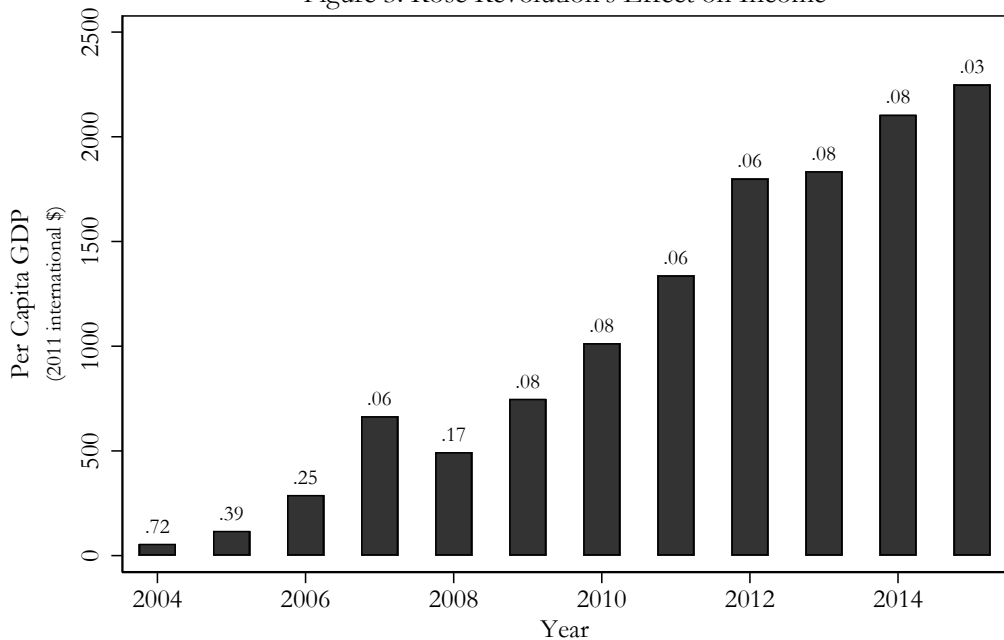
Source. Gwartney, Lawson, and Hall, 2016. World Bank Group, 2017. Transparency International, 2016. Authors' calculations.

Figure 2: Trends in Real per Capita GDP: Georgia and Synthetic Georgia



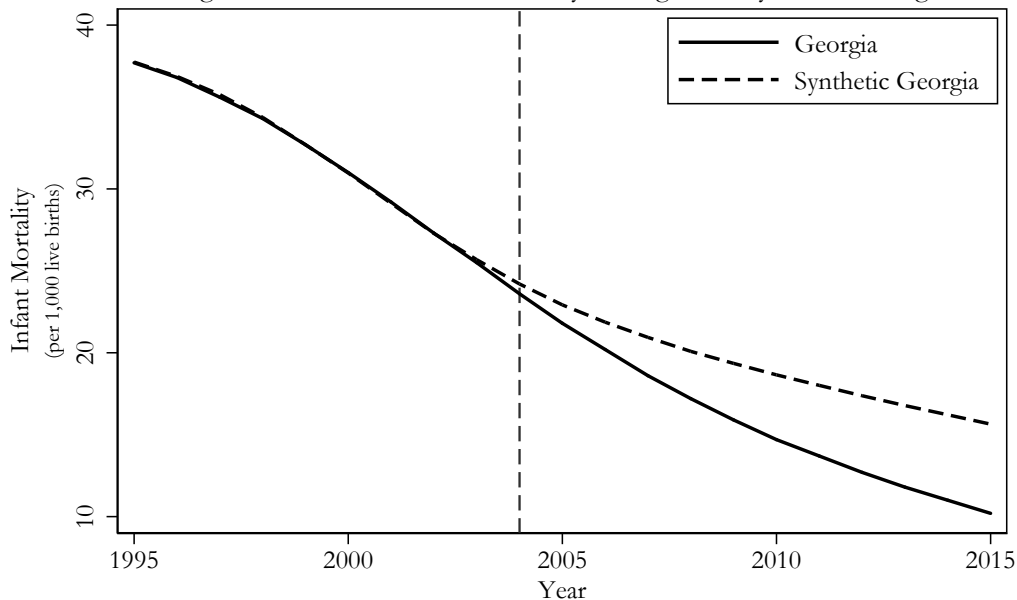
Note. The solid line is the time series for Georgia. The dashed line is the series for its synthetic counterpart. The years to the left of the vertical line are the pre-treatment years which are used in determining the synthetic control. The vertical line is the year 2004, the first year of the Rose Revolution treatment that we are studying. The gap between the solid and dashed lines to the right of the vertical line is our estimated causal effect of the Rose Revolution reforms

Figure 3: Rose Revolution's Effect on Income



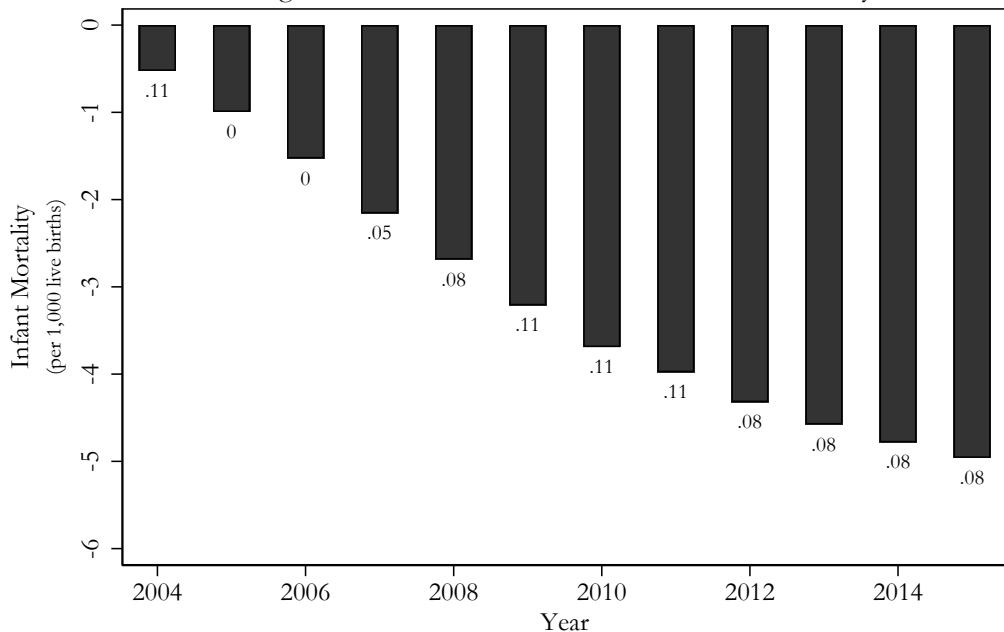
Note. The vertical bars show the distance between actual and synthetic Georgia for each year in the treatment period. The number at the top of each bar gives the p-value for the significance of the effect in that year.

Figure 4: Trends in Infant Mortality: Georgia and Synthetic Georgia



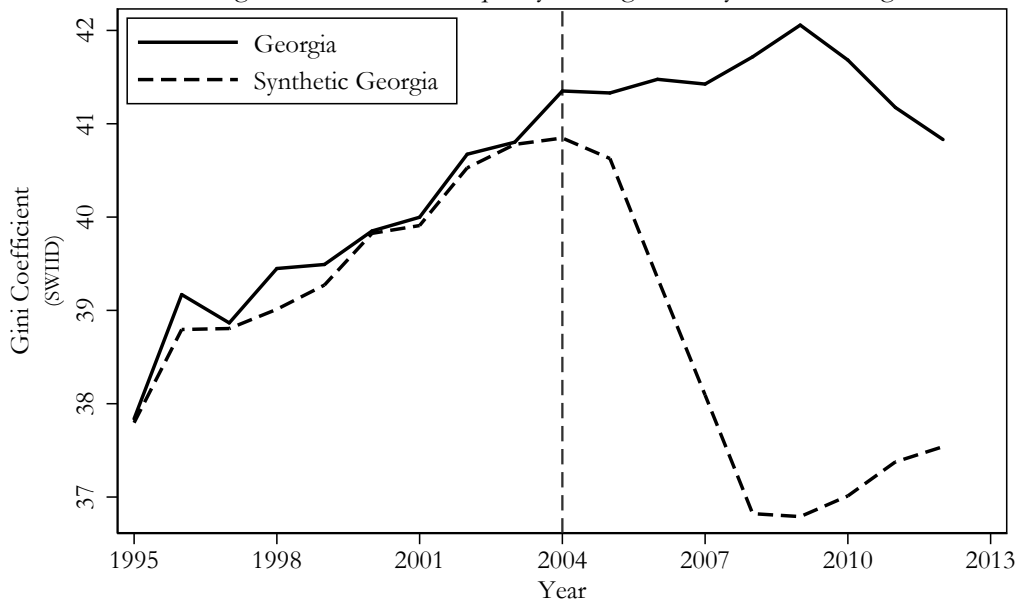
Note. The solid line is the time series for Georgia. The dashed line is the series for its synthetic counterpart. The years to the left of the vertical line are the pre-treatment years which are used in determining the synthetic control. The vertical line is the year 2004, the first year of the Rose Revolution treatment that we are studying. The gap between the solid and dashed lines to the right of the vertical line is our estimated causal effect of the Rose Revolution reforms

Figure 5: Rose Revolution's Effect on Infant Mortality



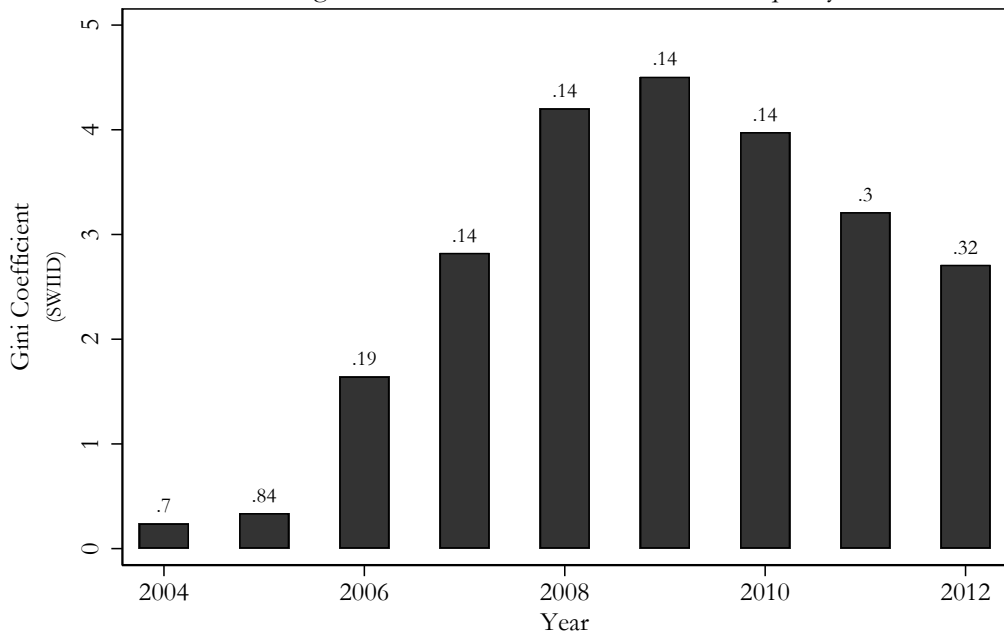
Note. The vertical bars show the distance between actual and synthetic Georgia for each year in the treatment period. The number at the top of each bar gives the p-value for the significance of the effect in that year.

Figure 6: Trends in Inequality: Georgia and Synthetic Georgia



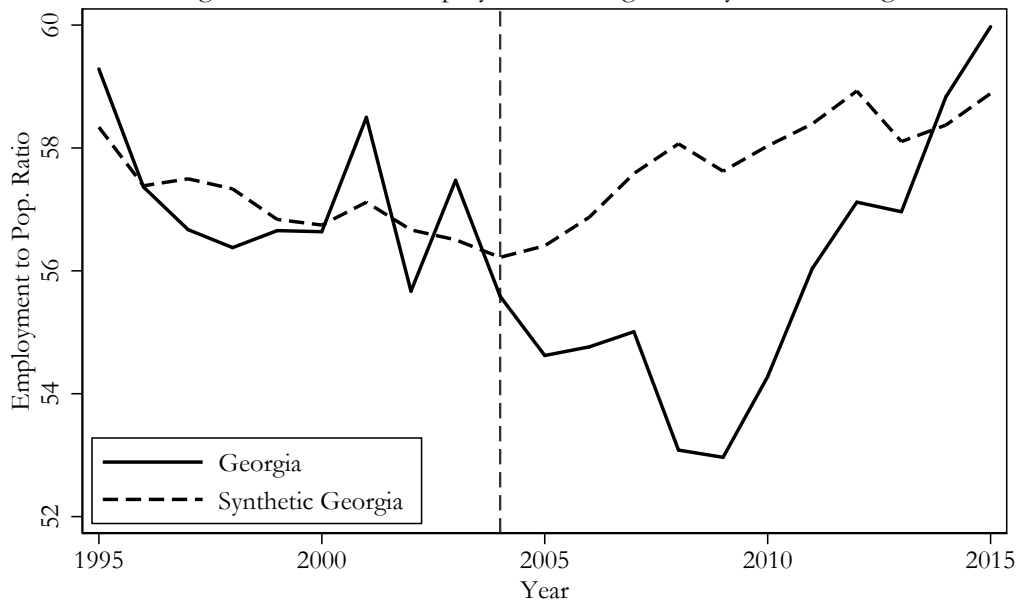
Note. The solid line is the time series for Georgia. The dashed line is the series for its synthetic counterpart. The years to the left of the vertical line are the pre-treatment years which are used in determining the synthetic control. The vertical line is the year 2004, the first year of the Rose Revolution treatment that we are studying. The gap between the solid and dashed lines to the right of the vertical line is our estimated causal effect of the Rose Revolution reforms

Figure 7: Rose Revolution's Effect on Inequality



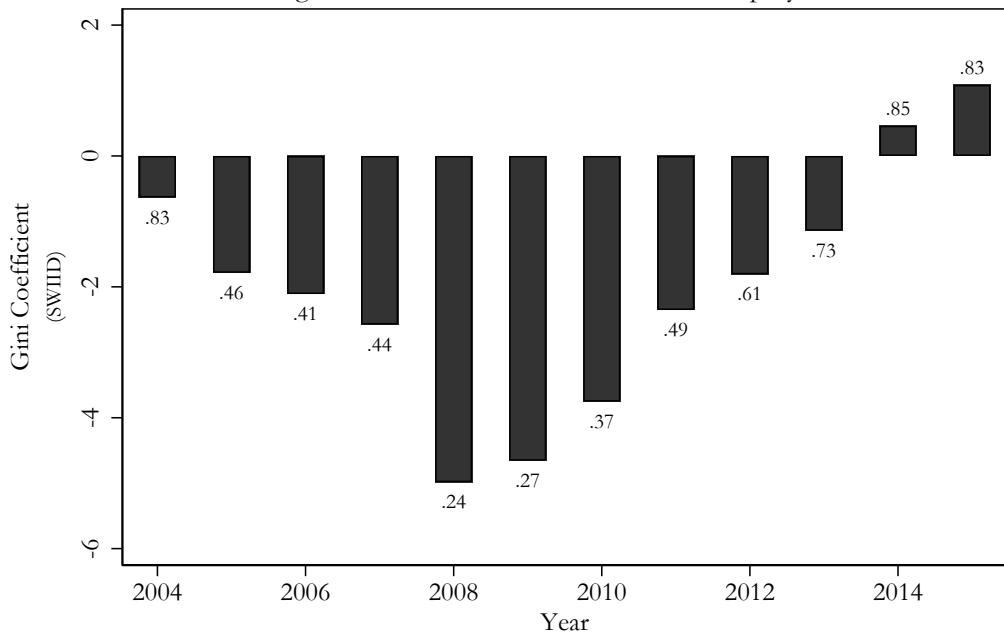
Note. The vertical bars show the distance between actual and synthetic Georgia for each year in the treatment period. The number at the top of each bar gives the p-value for the significance of the effect in that year.

Figure 8: Trends in Employment: Georgia and Synthetic Georgia



Note. The solid line is the time series for Georgia. The dashed line is the series for its synthetic counterpart. The years to the left of the vertical line are the pre-treatment years which are used in determining the synthetic control. The vertical line is the year 2004, the first year of the Rose Revolution treatment that we are studying. The gap between the solid and dashed lines to the right of the vertical line is our estimated causal effect of the Rose Revolution reforms

Figure 9: Rose Revolution's Effect on Employment



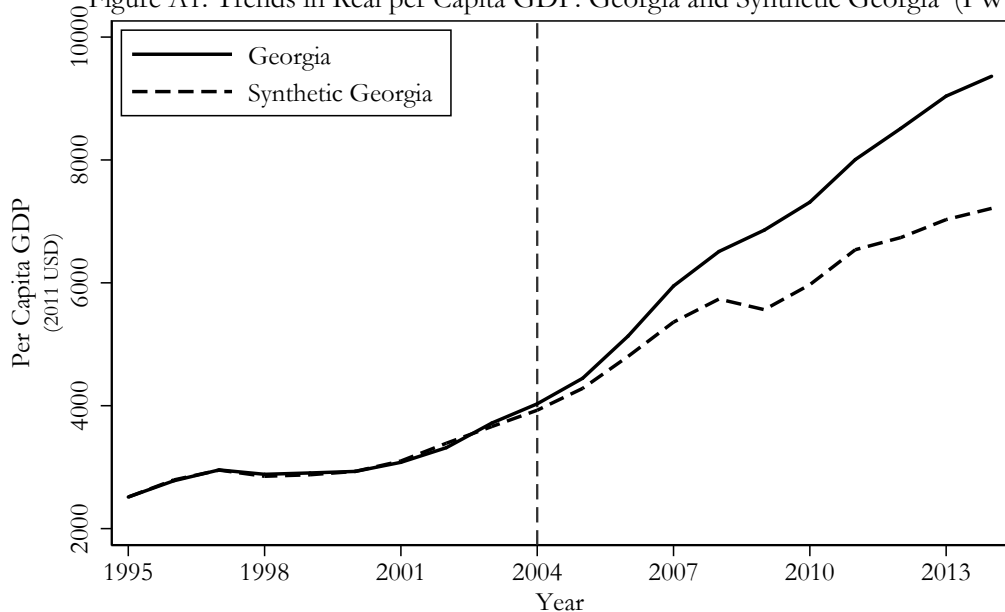
Note. The vertical bars show the distance between actual and synthetic Georgia for each year in the treatment period. The number at the top of each bar gives the p-value for the significance of the effect in that year.

TABLE A1: INCOME PREDICTOR MEANS (PENN WORLD TABLES)

Variables	Actual Georgia	Synthetic Georgia	Donor Pool Avg.
GDP per Capita (1995)	\$2,514.12	\$2,514.26	\$7,255.21
GDP per Capita (2000)	\$2,931.03	\$2,932.02	\$8,336.31
GDP per Capita (2003)	\$3,717.86	\$3,660.90	\$9,095.56
Government Consumption	0.21	0.29	0.27
Exports	0.04	0.10	0.17
Investment	0.15	0.13	0.18
Population Growth	-1.27	-0.47	0.53
RMSPE	--	35.067	5100.083

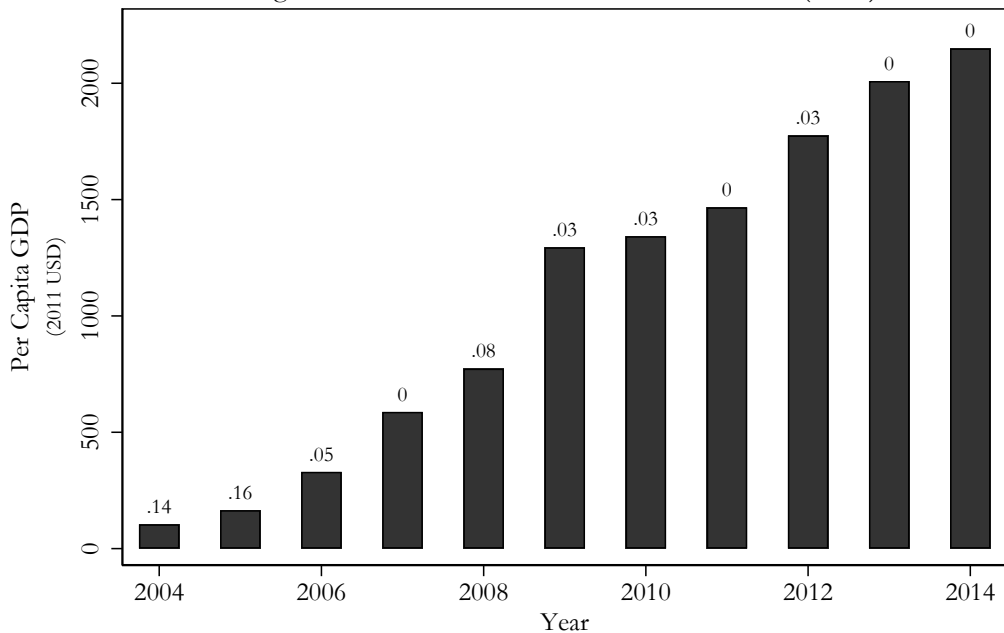
Note. This table shows mimics Table 3 from the analysis but uses Penn World Tables data, rather than World Bank data. We also include covariate values and RMSPE for the donor pool average to demonstrate the the superiority of the synthetic as a counterfactual. Unless otherwise indicated, variables are averaged across the pre-treatment period. The final row gives the root mean square prediction error for the unit of comparison.

Figure A1: Trends in Real per Capita GDP: Georgia and Synthetic Georgia (PWT)



Note. The solid line is the time series for Georgia. The dashed line is the series for its synthetic counterpart. The years to the left of the vertical line are the pre-treatment years which are used in determining the synthetic control. The vertical line is the year 2004, the first year of the Rose Revolution treatment that we are studying. The gap between the solid and dashed lines to the right of the vertical line is our estimated causal effect of the Rose Revolution reforms

Figure A2: Rose Revolution's Effect on Income (PWT)



Note. The vertical bars show the distance between actual and synthetic Georgia for each year in the treatment period. The number at the top of each bar gives the p-value for the significance of the effect in that year.